



Lab 11

MOSFET Diode and Triode Load

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11.1. Objective

To design and implement a common source amplifier with MOSFET triode-load and diode-load.

11.2. Pre-Lab Task

Task 1:

- i. Discuss in detail the difference between discrete and integrated Circuit design using MOSFET.

The main differences between discrete MOSFET circuit design and integrated MOSFET circuit design are summarized below:

1. Fabrication
 - Discrete: MOSFETs are manufactured as separate components and connected externally on a PCB.
 - Integrated: MOSFETs are fabricated together on a single chip using CMOS technology.
2. Size and Integration
 - Discrete: Larger in size with limited number of components.
 - Integrated: Very small (nanometer scale) with millions of transistors on one chip.
3. Design Approach
 - Discrete: Focus on circuit connections, voltage/current ratings, and thermal handling.
 - Integrated: Focus on transistor sizing, layout, and device-level effects.
4. Performance
 - Discrete: Higher parasitics, lower speed.
 - Integrated: Lower parasitics, higher speed and better precision.
5. Power Handling
 - Discrete: Suitable for high-power applications.
 - Integrated: Suitable for low-power, high-density applications.
6. Thermal Management
 - Discrete: Uses external cooling (heat sinks).
 - Integrated: Limited heat dissipation, handled through design optimization.
7. Cost and Scalability
 - Discrete: Cost increases with more components.
 - Integrated: High initial cost but very low cost per unit in mass production.